

1

Failure Pattern

(Don't overfit into one trial)



Problem

A one-off failure is noise; the same failure across seeds or tasks is a pattern worth a library entry. Without cross-trial comparison, repairs over-fit single trials.



When to Apply

- ≥ 2 trials exhibit the same symptom
- Failures cluster by object class, scene geometry, or action type



Strategy

Group attributed failures by (symptom, applicability) across trials. Recurring pairs become candidate library entries; stronger pairs are admitted as POS, NEG, or BLOCKED.

2

Trace Analysis

(robot execution engine assets)

3

Negative Constraint

(what not to do)

...



Failure Pattern

```

patterns = defaultdict(list)
for trial in attributed_trials:
    key = (trial.symptom,
           trial.applicability_class)
    patterns[key].append(trial)

for key, trials in patterns.items():
    if len(trials) >= 2:
        propose_library_entry(
            symptom=key[0], applies_to=key[1],
            evidence=trials
        )

```



Debugging